than auxiliary hypotheses. But as the saga of Neptune and Vulcan shows, it is far from obvious when a theory should be discarded in light of conflicting evidence, rather than rejecting some other assumption used to make the prediction. Claiming, then, that multiverse theories are unscientific because they are unfalsifiable is to reject them simply because they do not live up to an arbitrary standard that no scientific theory of any kind has ever met. Claiming that no data could ever force the rejection of a multiverse theory is merely stating that a multiverse theory is just like any other theory. And claiming that there could never be any observable evidence in favor of a multiverse theory is to forget Einstein's admonition that "it is the theory which decides what we can observe." Just as Grover Maxwell said in Chapter 8, what is considered observable can and does change over time, as scientific theories change. Atomic theory was once considered impossible to falsify, and atoms were once thought to be unobservable in principle. Evidence of a multiverse could share the same fate. Ultimately, arguments against a multiverse purportedly based on falsifiability are really arguments based on ignorance and taste: some physicists are unaware of the history and philosophy of their own field and find multiverse theories unpalatable. But that does not mean that multiverse theories are unscientific.

If scientific theories don't need to be falsifiable, what do they need to do? They need to give explanations, unify previously disparate concepts, and bear some relationship with the world around us. That's vague, of course, but science, like the people who do it and the world it describes, is complicated. Pat answers to complex questions, like Popper's cry of "falsifiability!", should always be suspicious: as H. L. Mencken once said, "There is always a well-known solution to every human problem—neat, plausible, and wrong."

What, then, is the right solution to the human problem of the Copenhagen interpretation? For, despite everything—despite pilot waves and many worlds, despite Bell and Bohm and Everett, despite the rise of quantum computing and the fall of logical positivism—Copenhagen is still dominant within physics. Copenhagen is still the view taught in every